Name:_____Block:_____

Algebra 1 Unit 1

Real Numbers, Expressions & Equations ≅ 8 Days

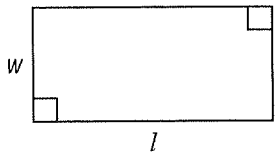
Real Number System • GCF and LCM of Numbers • Simplifying Radicals • Properties of Real Numbers
Simplifying and Evaluating Expressions
Combining Like Terms and Distributive Property
Translating Expressions
Equations: 1 step, 2 step, multi-step, absolute value, literal equations, solutions of equations, proportions
Properties of equality

Is the number real or irrational? SIMPLIFY IT! $\sqrt{12}$	**What is the GCF of 24 and 16?** **What is the LCM of 24 and 16?**	**Simplify the expression:** $8x - 2 + x - 3$	**Simplify the expression:** $-4(2x - 5)$
SImplify: $\sqrt{25} + \left\|(8 \div 2)^2\right\|$	**Solve the equation:** $\frac{2}{3}x - 7 = 10$	**Solve the equation:** $\|3x - 5\| = 10$	

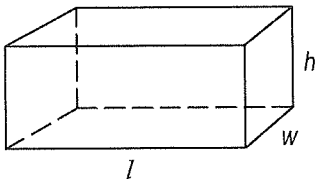
Start Date:_____Scheduled Test Date:_____

Algebra I Formula Sheet

Formulas that you may need to solve questions on this exam are found below.
You may use calculator π or the number 3.14.



$A = lw$



$V = lwh$

Linear Equations

Slope: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

Point-Slope Formula: $(y - y_1) = m(x - x_1)$

Slope-Intercept Formula: $y = mx + b$

Standard Equation of a Line: $Ax + By = C$

Arithmetic Properties

Additive Inverse: $a + (-a) = 0$

Multiplicative Inverse: $a \cdot \dfrac{1}{a} = 1$

Commutative Property: $a + b = b + a$
$a \cdot b = b \cdot a$

Associative Property: $(a + b) + c = a + (b + c)$
$(a \cdot b) \cdot c = a \cdot (b \cdot c)$

Identity Property: $a + 0 = a$
$a \cdot 1 = a$

Distributive Property: $a \cdot (b + c) = a \cdot b + a \cdot c$

Multiplicative Property of Zero: $a \cdot 0 = 0$

Additive Property of Equality:
If $a = b$, then $a + c = b + c$

Multiplicative Property of Equality:
If $a = b$, then $a \cdot c = b \cdot c$

Copyright © 2011 by the Pennsylvania Department of Education. The materials contained in this publication may be duplicated by Pennsylvania educators for local classroom use. This permission does not extend to the duplication of materials for commercial use.

The Real Number System

REAL NUMBERS

____ = _____

Decimals that never end and have no repeating pattern. *Think ***MUMBO JUMBO NUMBERS***

IRRATIONAL NUMBERS

____ = _____

All fractions and decimals that end or have a repeated pattern.

RATIONAL NUMBERS

____ = _____

INTEGERS

____ = _____

WHOLE NUMBERS

____ = _____

NATURAL NUMBERS

____ = _____

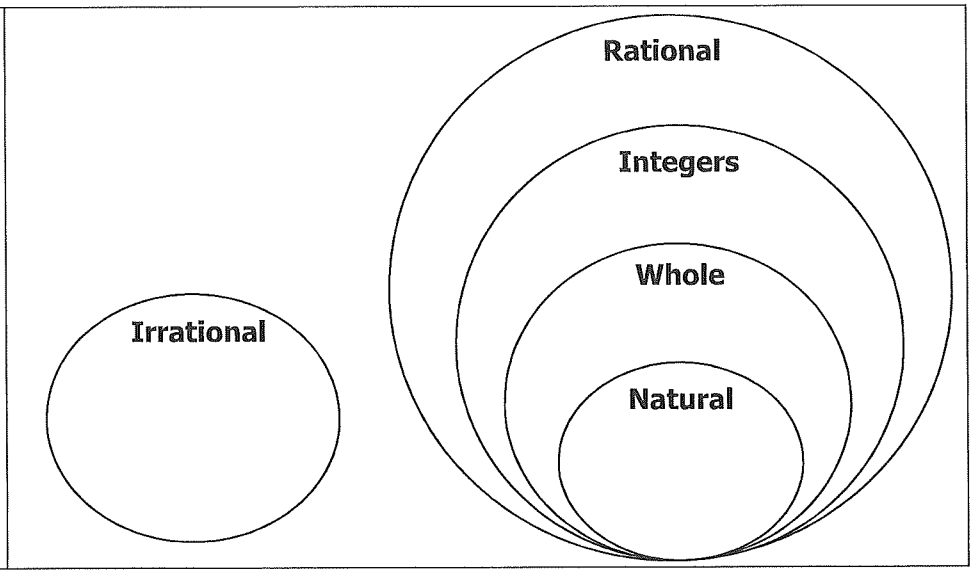
Directions: Name all sets of numbers to which each number belongs.		
1. 30	**2.** -11	**3.** $5\frac{4}{7}$
4. $\sqrt{21}$	**5.** 0	**6.** $-\sqrt{9}$
7. $\frac{6}{3}$	**8.** π	**9.** $5.\overline{3}$
10. $\frac{\sqrt{144}}{-3}$	**11.** –2.875	**12.** $\frac{\sqrt{17}}{\sqrt{17}}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Organizing the Real Numbers

Directions: Place the LETTER of each value in its location in the real number system below.

A. 7	**B.** $-\frac{36}{12}$
C. $-\sqrt{36}$	**D.** π
E. 6.48	**F.** -2
G. $-\sqrt{10}$	**H.** 0
I. $2\frac{1}{6}$	**J.** $-0.\overline{7}$
K. $\sqrt{\frac{4}{81}}$	



1. Which expression does **not** name an integer?

A. -15 **C.** 0

B. $\frac{6}{14}$ **D.** $\frac{12}{6}$

2. Which expression represents an irrational number?

A. $0.\overline{18}$ **C.** $\frac{2}{3}$

B. $\sqrt{75}$ **D.** $\sqrt{3}-\sqrt{3}$

3. Which number is **not** a whole number?

A. 8 **C.** 0

B. -10 **D.** $\frac{18}{3}$

4. Which of the following is a true statement?

A. -9 is a whole number **C.** 0 is a natural number

B. $\sqrt{25}$ is an irrational number **D.** $\frac{2}{3}$ is a rational number

5. Which of the following statements is false?

A. All real numbers are rational numbers. **C.** All natural numbers are integers.

B. Every integer is a rational number. **D.** Every whole number is a real number.

6. Give an example of a number that is a whole number, but not a natural number. _____

7. Give an example of a real number that is not rational. _____

8. Give an example of a rational number that is not an integer. _____

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 1:** The Real Numbers

Directions: Name all sets of numbers to which each real number belongs.

1. 12 **2.** -15 **3.** $1\frac{1}{2}$

4. 3.18 **5.** $\sqrt{48}$ **6.** $9.\overline{3}$

7. $-2\frac{7}{9}$ **8.** $\sqrt{25}$ **9.** $\sqrt{3}$

10. $-\sqrt{64}$ **11.** $-\sqrt{12}$ **12.** $\frac{8}{4}$

Directions: Place the LETTER of each value in its location in the real number system below.

A. 2.125 **B.** 0

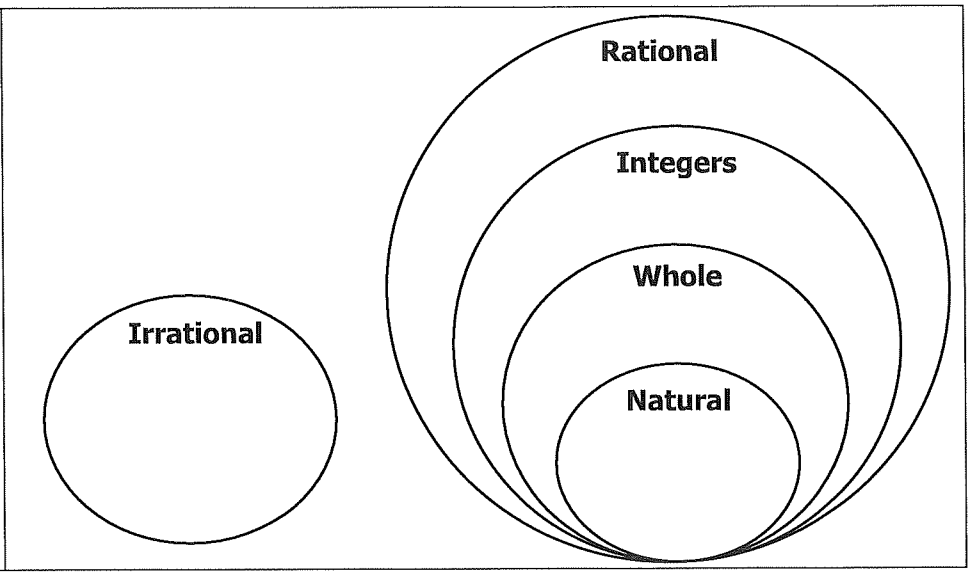
C. π **D.** $-\frac{15}{5}$

E. $3\frac{1}{4}$ **F.** $0.\overline{4}$

G. 23 **H.** -1

I. $\sqrt{\frac{16}{25}}$ **J.** $-\sqrt{49}$

K. $-\sqrt{35}$



Questions:

1. Give an example of a whole number that is not a natural number. _____

2. Give an example of a rational number that is not a whole number. _____

3. Give an example of a real number that is irrational. _____

4. Are integers always/sometimes/never whole numbers? _____

5. Are natural numbers always/sometimes/never rational numbers? _____

6. Are irrational numbers always/sometimes/never rational numbers? _____

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

LCM and GCF Practice Name: _____

Directions: Show your work to answer the questions. Use your notes to help you.

FIND THE **LCM** OF 6 AND 3	FIND THE **LCM** OF 12 AND 5	FIND THE **GCF** OF 72 AND 42	FIND THE **GCF** OF 37 AND 24
FIND THE **LCM** OF 4 AND 10	FIND THE **LCM** OF 2 AND 7	FIND THE **GCF** OF 21 AND 63	FIND THE **GCF** OF 15 AND 60
FIND THE **LCM** OF 3 AND 5	FIND THE **LCM** OF 7 AND 8	FIND THE **GCF** OF 100 AND 25	FIND THE **GCF** OF 24 AND 96

H

Simplifying Radicals

1) Complete the Perfect Square Root Chart →

2) To simplify a non-perfect square root, follow these steps:

$$\sqrt{128}$$

STEP ONE: Find perfect squares that the number inside the radical are divisible by. Write as separate radicals.

STEP TWO: Find the square root of each perfect square and keep these outside the radical.

STEP THREE: Simplify by multiplying all numbers outside the radical to find your coefficient.

$12 = \sqrt{\quad}$
$11 = \sqrt{\quad}$
$10 = \sqrt{\quad}$
$9 = \sqrt{\quad}$
$8 = \sqrt{\quad}$
$7 = \sqrt{\quad}$
$6 = \sqrt{\quad}$
$5 = \sqrt{\quad}$
$4 = \sqrt{\quad}$
$3 = \sqrt{\quad}$
$2 = \sqrt{\quad}$
$1 = \sqrt{\quad}$

a) $\sqrt{90}$ b) $\sqrt{108}$ c) $\sqrt{150}$

Name: _____ Date: _____

Notes: Properties of Real Numbers

Do Now: Solve the following equation for x.

$$6 + 5(7 + 2x) = 8x - 13$$

Match the properties on the left with their corresponding math on the right.

_____ Commutative Property of Addition	A) $a \cdot 0 = 0$
_____ Associative Property of Addition	B) $a + (b + c) = (a + b) + c$
_____ Commutative Property of Multiplication	C) $a(b + c) = ab + ac$
_____ Associative Property of Multiplication	D) $a + b = b + a$
_____ Distributive Property	E) $a \cdot 1 = a$
_____ Identity Property of Addition	F) $a + 0 = a$
_____ Identity Property of Multiplication	G) $(ab)c = a(bc)$
_____ Zero Property of Multiplication	H) $ab = ba$

1) **Commutative:** Numbers can be added or multiplied in any order.

\+ ×

2) **Associative:** Add or multiply regardless of how the numbers are grouped.

\+ ×

3) **Identity:** When you add/multiply to obtain the same number.

\+ ×

4) **Inverse:** When you add/multiply to obtain the identity.

$+$ $\times$

5) **Distributive:** Multiply a sum or difference by multiplying each addend separately and then add the products.

6) **Zero Property:** Any value multiplied by 0 has a product of 0.

7) **Addition Property of Equality:** Add the same quantity to both sides of the equation.

8) **Subtraction Property of Equality:** Subtract the same quantity to both sides of the equation.

9) **Multiplication Property of Equality:** Multiply the same quantity to both sides of the equation.

10) **Division Property of Equality:** Divide the same quantity to both sides of the equation.

Practice:

1. When solving the equation $5(x-6)+5=9$, Mary wrote $5(x-6)=4$ as her first step. Which property justifies Mary's first step?

(1) subtraction property of equality
(2) distributive property of multiplication over subtraction
(3) associative property
(4) multiplication property of equality

2. A part of Bianca's work to solve the equation $3(7x^2 - 10) = 15x^2 - 8x$ is shown below

Given: $3(7x^2 - 10) = 15x^2 - 8x$
Step 1: $21x^2 - 30 = 15x^2 - 8x$

What property did Bianca use to obtain step 1?
(1) addition property of equality
(2) distributive property of multiplication over subtraction
(3) associative property
(4) multiplication property of equality

3 Given $\blacksquare \neq 0$, the equation where $\nabla(\blacksquare) = \blacksquare$ is an example of the
(1) associative property
(2) inverse property
(3) identity property
(4) zero property

4. When solving for the value of x in the equation $-5(3x - 9) + 5 = 65$, Guillermo wrote the following lines on his paper.

[line 1] $-5(3x - 9) + 5 = 65$
[line 2] $-5(3x - 9) = 60$
[line 3] $-15x - 45 = 60$
[line 4] $-15x = 105$
[line 5] $x = -7$

Between which two lines did Guillermo make a mistake using a property, which resulted in the incorrect answer?
(1) line 1 – line 2
(2) line 2 – line 3
(3) line 3 – line 4
(4) line 4 – line 5

5. Which equation illustrates the additive inverse property?

(1) $\Omega\left(\frac{1}{\Omega}\right) = 1$
(2) $\Omega + (-\Omega) = 0$
(3) $\Omega + \Omega = 2\Omega$
(4) $\Omega(\Omega) = \Omega^2$

6. The equation $\vartheta(\blacksquare + \Delta) = \vartheta\blacksquare + \vartheta\Delta$ is an example of the
(1) associative property
(2) commutative property
(3) distributive property
(4) identify property of multiplication

7. Juliane is solving the equation $2(3x - 9) = -6$ is shown below. Identify the property used to obtain each of the steps.

$2(3x - 9) = -6$

$6x - 18 = -6$ _____

$6x = 12$ _____

$x = 2$ _____

8. When solving the equation $(5 + 7) + 10 = x$, Ben rewrote the equation $5 + (7 + 10) = x$ as his first step. Which property justifies Ben's first step?
(1) addition property of equality
(2) distributive property of multiplication over addition
(3) associative property
(4) commutative property

9. When solving for the value of x in the equation $-6(x - 5) + 2(4x + 3) = 12$, Layla wrote the following lines on his paper.

[line 1] $-6x + 30 + 8x + 6 = 12$
[line 2] $2x + 36 = 12$
[line 3] $2x = 48$
[line 4] $x = 24$

Which property did Layla perform incorrectly, which resulted in the incorrect answer?
(1) line 1 – line 2
(2) line 2 – line 3
(3) line 3 – line 4
(4) Laya did not make a mistake.

Name: _____ Date: _____

Homework: Properties of Real Numbers

1 Janita simplified an equation to $-b + b = 0$. Which property of real numbers is shown by this equation?

(1) additive identity property
(2) multiplicative identity property
(3) additive inverse property
(4) multiplicative inverse property

2 Given the following equations:

I. $a + b = b + a$
II. $a + (b + c) = (a + b) + c$
III. $5a(8a^2) = 8a^2(5a)$
IV. $ab = ba$

Which equations(s) represent the commutative property?
(1) I and IV, only
(2) II, only
(3) I, II, III, and IV
(4) I, III, and IV

3 Brittany is solving the equation $(7 - 4x)2 = -34$ is shown below. Identify the property used to obtain each of the steps.

$(7 - 4x)2 = -34$

$2(7 - 4x) = -34$ _____

$14 - 8x = -34$ _____

$-8x = -48$ _____

$x = 6$ _____

4 When solving the equation $7(2x^2 - 4) - 11 = 5x^2 - 2$, Emilia rewrote the equation $7(2x^2 - 4) = 5x^2 + 9$ as her first step. Which property justifies Emilia's first step?
(1) distributive property of multiplication over subtraction
(2) addition property of equality

(3) associative property
(4) division property of equality

5 State whether each statement is *true* or *false*. If *true*, state the property of real numbers that proves the statement true. If *false*, give a counterexample.

a For all real numbers x and y, $x(y) = y + x$.

b For all real numbers a, $a + 0 = a$.

c For all real numbers x, y, and z, $x(y + z) = xy + xz$.

6 The equation $\sqcup\sqcap = \sqcap\sqcup$ is an example of the
(1) associative property
(2) commutative property
(3) distributive property
(4) identity property of multiplication

7 Which equation correctly illustrates the distributive property?

(1) $w(x) = x(w)$
(2) $-w(x - y) = -wx - wy$
(3) $w(x + y) = wx - wy$
(4) $-w(-x - y) = wx + wy$

Properties Warm-Up

_____ **1.** $16 = 16$

_____ **2.** $7 + (a + b) = (7 + a) + b$

_____ **3.** $(4y + 1) \cdot 0 = 0$

_____ **4.** If $5 \cdot 2 = 10$ and $10 = \sqrt{100}$, then $5 \cdot 2 = \sqrt{100}$

_____ **5.** $-2(x - y) = -2x + 2y$

_____ **6.** $3 \cdot \frac{1}{3} = 1$

_____ **7.** If $8x = 24$, then $24 = 8x$

_____ **8.** $(2 + 1) + 9 = (1 + 2) + 9$

_____ **9.** $6m \cdot 1 = 6m$

A. Commutative Property

B. Associative Property

C. Distributive Property

D. Identity Property

E. Inverse Property

F. Property of Zero

G. Reflexive Property

H. Symmetric Property

I. Transitive Property

10. True/False: Negative numbers closed under subtraction. Provide an example if false.

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Absolute Value	
Examples	**Directions:** Evaluate each expression.

1. $\lvert -13 \rvert$ =	**2.** $\lvert 21 \rvert$ =	**3.** $\lvert -3 \rvert + \lvert -5 \rvert$ =
4. $\lvert 9 \rvert + \lvert -8 \rvert$ =	**5.** $\lvert -12 \rvert + \lvert 15 \rvert$ =	**6.** $\lvert 21 - 17 \rvert$ =
7. $\lvert -11 \rvert - \lvert -5 \rvert$ =	**8.** $\lvert 4 \rvert - \lvert -4 \rvert$ =	**9.** $\lvert 15 - 23 \rvert$ =
10. $\lvert 24 \rvert - \lvert -15 \rvert$ =	**11.** $\lvert 28 \rvert - \lvert -26 \rvert$ =	**12.** $\lvert -3 \rvert - \lvert 19 \rvert$ =

Order of Operations	

Examples	**Directions:** Evaluate each expression.
13. $7 + 54 \div 3(2)$	**14.** $17 - 2^5 \div 4 + 6$
15. $24 - 4^2 \cdot 3 + 15$	**16.** $\dfrac{(7 - 2^2) + 17}{-14 + 2 \cdot 5}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	17. $(8-5)^3-\lvert 5^2-4^3\rvert \div 3$	**18.** $\lvert -7^2+3^2\cdot 4\rvert +18\div 2\cdot 5$
	19. $\dfrac{5^2-3^4}{(2^3-5)\cdot(6^2\div 9)}$	**20.** $-\lvert -9-10\rvert +12^2\div 6\cdot 3$
	21. $20^2\div\{3(5-9)^2+2\}$	**22.** $\dfrac{12\left[30-(9+4^2)\right]}{\lvert 10\rvert -\lvert -6\rvert}$
	23. $\dfrac{84}{12}+(5-7^2)-72\div 6\cdot 2$	**24.** $\dfrac{-(8-11)^3-3\lvert -14\rvert}{2^4-7}$

Summary: _____

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 4:** Order of Operations

Directions: Evaluate each expression using the order of operations.	
1. $17-5\cdot 4\div 2$	**2.** $40-32\div 8+5\cdot -2$
3. $35-14\div 2+8^2$	**4.** $(4-7)^2-6\cdot 7+20$
5. $2[18-(5+3^2)\div 7]$	**6.** $[(-5+1)\div 2]^3-\lvert -7\rvert$
7. $1+(-2-5)^2+(14-17)\cdot 4$	**8.** $\dfrac{6(2+4)-1}{2\cdot 3+1}$
9. $\dfrac{2^5\div 8+3}{4+3}$	**10.** $\dfrac{17\cdot 5-3\cdot 5}{3^2+1}$
11. $\dfrac{13+7^2\div 7}{9-20\div 4-16}$	**12.** $-\lvert -16\rvert \div \dfrac{(3^2-4)}{5-10}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
ALGEBRAIC EXPRESSION	
SUBSTITUTION *Property*	If _____, then _____
EVALUATING *Expressions*	**To evaluate an expression with variable replacements:** • _____ the variables with their given values. • Each time you substitute a variable with a number, put _____ around the number! • Follow the _____ _____ _____ to evaluate!
EXAMPLES	**Directions:** Evaluate each expression using the variable replacements. **1.** $ab^2 + c$ if $a = 2$, $b = 4$, $c = 7$ **2.** $3x^2 - 4x$ if $x = -2$
YOU TRY!	**Directions:** Evaluate each expression using the variable replacements. **3.** $a^2b - b^2$ if $a = 3$ and $b = -4$ **4.** $a^2b - b^2$ if $a = 4$ and $b = -7$ **5.** $-y^2 - 3xy$ if $x = -4$ and $y = 2$ **6.** $-y^2 - 3xy$ if $x = -\frac{5}{6}$ and $y = -12$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	7. $\frac{a^2+b^2}{a-b}$ if a = 3 and b = 4	**8.** $\frac{a^2+b^2}{a-b}$ if a = -3 and b = -5
	9. $x^2-2(x-y)-z^3$ if x = 2, y = -2, and z = -3	**10.** $x^2-2(x-y)-z^3$ if x = -3, y = 4 and z = -1
	11. $-a^2-2bc-\lvert c\rvert$ if a = -2, b = 3, and c = -3	**12.** $-a^2-2bc-\lvert c\rvert$ if a = -3, b = -5, and c = 2

Summary: _____

_____

_____

_____

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Parts of an Expression	$7x + 3 - 5x + 8 + 4x - 1$ **Variable Terms** (Terms WITH a variable) **Constant Terms** (Terms WITHOUT a variable) **Coefficients** (Number NEXT TO a variable)
Combining Like Terms	You can simplify an algebraic expression by **combining like terms.** This means to combine common variable terms and constant terms. **Example:** Simplify the expression below: $7x + 3 - 5x + 8 + 4x - 1 =$ _____
Examples	**Directions:** Simplify each expression.

1. $3x + 6x$	**2.** $9a - 10a$
3. $7m - 5m - 6$	**4.** $9 - 6w + 5$
5. $7p - 1 - 9p + 5$	**6.** $5h - 6 - 8 + 7h$
7. $5x + 7y - x + 2y + 3x$	**8.** $26a + 2b - 4a + 10b$

© Gina Wilson (All Things Algebra®, LLC), 2012 2016

with Exponents

Terms with the same exponents or combination of variables and exponents are considered like terms. When combined, be sure to NOT change their exponents!

Examples

9. $5x^2 + 7x^2$

10. $-15n^5 + 4n^5$

11. $8c^3 - 7c^3 + 16c^3$

12. $8k^2 - k - 5k + 7 - 2k^2$

13. $-4a^2b + 7 + 10a^2b$

14. $x^2 + 2xy - 7xy + 4y^2$

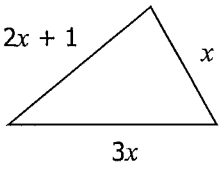
15. $a^2 + 7ab - 10ab + b^2$

16. $4x^2 - 7 + 2x - 3x^2 + 8x - 6$

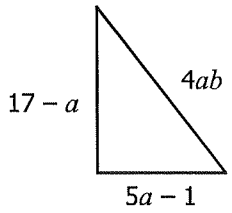
Geometric Applications

Directions: Give the perimeter of each figure as a simplified expression.

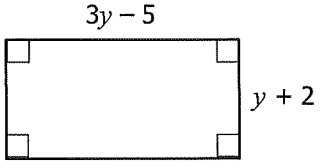
17.



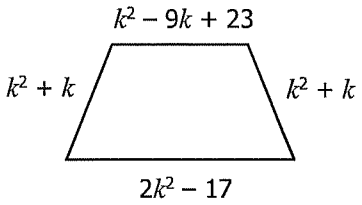
18.



19.



20.



© Gina Wilson (All Things Algebra®, LLC), 2012 2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Distributive Property **Recall:** $a(b+c) =$ or $a(b-c) =$	**1.** $7(x + 4)$	**2.** $2(b - 3)$	**3.** $-4(y + 3)$
	4. $-5(m - 2)$	**5.** $-(y - 9)$	**6.** $8(4 - b)$
	7. $-4(-w - 10)$	**8.** $5(2m - 3)$	**9.** $-3(2x - 4)$
	10. $3(m + n)$	**11.** $x(y + 4)$	**12.** $d(c - 4)$
	13. $6a(b + 2c)$	**14.** $2(x - y + 8)$	**15.** $x(2y - 5z + 1)$
	16. $\frac{1}{4}(8x - 4)$	**17.** $\frac{1}{6}\left(\frac{3}{5}x + 18\right)$	**18.** $-\frac{1}{8}\left(\frac{4}{5}x - 24\right)$
Distribute AND Combine!	To **simplify an expression** means to ensure there are **no parentheses** and **no like terms**. In order to do this, distribute first (if needed),then combine like terms.		
Examples	**19.** $8(2x - 3) - 6x$	**20.** $9(2k - 4) - 2(7k - 12)$	
	21. $10 - (y - 6) - y$	**22.** $6 + 8(4w - 7) - (2w + 1)$	
	23. $\frac{9}{4}\left(\frac{10}{3}x - 4\right) - \frac{1}{4}(2x - 8)$	**24.** $\frac{7}{8}\left(\frac{4}{5}x + 16\right) + \frac{3}{5}\left(\frac{1}{2}x - 10\right)$	

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Key Words & Phrases

ADDITION +

- _____
- _____
- _____
- _____
- _____

SUBTRACTION −

- _____
- _____
- _____
- _____
- _____

MULTIPLICATION ●

- _____
- _____
- _____
- _____
- _____

DIVISION /

- _____
- _____
- _____
- _____
- _____

(Parentheses Phrases)

- _____
- _____

"Turn Around" Phrases

- _____
- _____

Examples

Directions: Translate each expression or equation.

1. "the product of a number and 3"

2. "the difference between a number and 10"

3. "the quotient of a number and -5"

4. "four subtracted from a number"

5. "the sum of twice a number and 7"

6. "the product of a number and 10, increased by 1"

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	7. "four less than a number squared"	**8.** "twice a number divided by 5"
Equations	**9.** "the sum of twice a number and 10 is 36"	**10.** "three times the difference of a number and 7 is 18"
	11. "four times a number divided by 8 is 3"	**12.** "five times the sum of a number and 6 is 48"
	13. "one half of a number is 8 less than the number itself"	**14.** "the product of a number and 3 is the same as the sum of that number and six"
	15. "fifteen decreased by the quotient of a number and 3 is 20"	**16.** "four less than the product of a number and 7 is eight more than that number"

Inequalities	**LESS THAN**	**LESS THAN OR EQUAL TO**	**GREATER THAN**	**GREATER THAN OR EQUAL TO**

Examples	**17.** "a number is less than -10"	**18.** "a number is less than or equal to 8"
	19. "a number is greater than or equal to -4"	**20.** "a number is at most 45"
	21. "a speed limit of 55 mph"	**22.** "a number is at least 16"
	23. "a number is no less than 10"	**24.** "a number is no more than 26"
	25. "a maximum number of 25"	**26.** "a minimum number of 5"

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
One-Step Equations	**1.** $m + 12 = 10$	**2.** $-2 = g - 9$
	3. $-7y = -91$	**4.** $\frac{a}{9} = -4$
Fractions To "get rid" of a fraction, multiply by the _____!	**5.** $\frac{2}{3}x = 10$	**6.** $\frac{4}{9}w = -8$
	7. $-\frac{6}{5}k = 12$	**8.** $-\frac{1}{2}m = -9$
Two-Step Equations	**To Solve a Two-Step Equation:** 1. Undo the **Addition/Subtraction** (to remove constant term) 2. Undo the **Multiplication/Division** (to remove coefficient)	
	9. $6x + 8 = 50$	**10.** $2n - 5 = 11$
	11. $13 = -4k + 9$	**12.** $7 - 3y = 34$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	13. $\frac{x}{2} - 7 = 9$	**14.** $11 = \frac{c}{-5} + 8$
	15. $\frac{3}{5}x + 22 = 28$	**16.** $-\frac{1}{3}m + 1 = -7$
	17. $-10 + \frac{7}{4}p = -38$	**18.** $15 = 9 - \frac{1}{2}x$
Watch Out!	**The examples below are different in that the multiplication/division is done FIRST, followed by the addition/subtraction.**	
	19. $\frac{x+11}{8} = -3$	**20.** $\frac{n-5}{-2} = -7$
	21. $1 = \frac{a-13}{-6}$	**22.** $4 = \frac{w+8}{9}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 10:** One & Two-Step Equations

Directions: Solve each equation. **SHOW ALL STEPS!**		
1. $y - 17 = -37$	**2.** $-12 + k = 18$	**3.** $5m = -20$
4. $\frac{x}{9} = 3$	**5.** $\frac{4}{5}z = 16$	**6.** $3p - 4 = 11$
7. $-5a + 7 = -38$	**8.** $31 - 12n = 211$	**9.** $\frac{x}{3} - 15 = -7$
10. $\frac{f}{-4} + 11 = 5$	**11.** $\frac{4}{3}v + 25 = 33$	**12.** $\frac{5}{8}c + 13 = 18$
13. $-\frac{2}{3}w + 53 = 59$	**14.** $11 = \frac{4}{5}h - 1$	**15.** $-\frac{3}{8}r + 41 = -22$
16. $34 = 28 - \frac{2}{5}y$	**17.** $\frac{n+4}{-2} = -9$	**18.** $-5 = \frac{x-1}{8}$

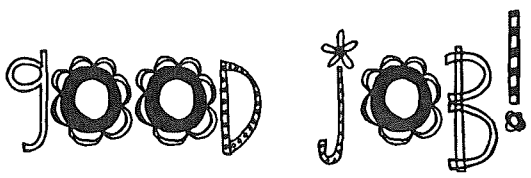
© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Translate... and Solve!

1 "Twice a number plus 4 is 8"		2 "The product of 4 and a number plus 17 is 5."	
3 "Twice a number minus 5 is 7"		4 "The quotient of a number and 6 minus 3 is 1"	
5 "Two more than 4 times a number is -18"		6 "Seven less than twice a number is 5"	
7 "Five less than the quotient of a number and 3 is -7"		8 "One subtracted from the product of 4 and a number is 11"	
9 "The quotient of a number and -9, increased by 10 is 11"		10 "Half of a number decreased by 8 is -3"	

© Gina Wilson (All Things Algebra®, LLC), 2012 2016

11 "The sum of three-fourths of a number and 1 is -5"		12 "Ten minus one-third of a number is 4"	
13 "Four more than the quotient of a number and 3 is at least 9"		14 "Negative three times a number increased by seven is less than -11"	
15 "Seven more than one-sixth of a number is greater than three"		16 "The sum of 5 and the twice a number is at most 27"	
17 "Four less than two-fifths of a number is no less than 2"		18 "Three-fourths of a number decreased by 10 is greater than or equal 5"	
19 "The quotient of a number and -5, increased by 7 is at minimum 1"		20 "One less than half a number is a maximum of 19"	



© Gina Wilson (All Things Algebra®, LLC), 2012 2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Steps to Solve a Multi-Step Equation	
Examples	**1.** $9x + 1 - 7x - 5 = -20$
	2. $91 = -7(3a - 1)$
	3. $4m - 5(3m + 10) = 126$
	4. $-3(k - 8) - (k + 5) = 23$
	5. $20 = 10x - 6(2x + 5)$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	6. $8(2w - 1) - 4w = -116$
	7. $11h - (2h - 1) = 118$
	8. $-25 = \frac{1}{2}(10x - 2) + 3x$
	9. $7 - \frac{5}{2}(8r - 6) + 2r = 32$
Translate & Solve 	**10.** *"Five times the difference of twice a number and three, decreased by the sum of the number and eight, equals 13."*

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 1:** Multi-Step Equations

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $-4x - 7 - 3x + 4 = 25$	**2.** $-4(7a + 5) = -160$
3. $3k + 2(5k - 3) = 7$	**4.** $-79 = 7w + 3(4w - 1)$
5. $9 - 4(2p - 1) = 41$	**6.** $y - 3(2y - 7) = 76$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

7. $15 - 5(4c - 7) = 50$	**8.** $-(6h + 7) + 8 = -19$
9. $-\frac{1}{3}(9x + 42) - 5x = -70$	**10.** $8 = -18 + \frac{3}{8}(16 - 40n)$
11. $2(5v + 3) - 4(7v + 1) = -88$	**12.** $7(3r - 1) - (r + 5) = -52$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Steps to Solve		
Examples	**1.** $5y - 8 = 3y + 12$	**2.** $-6x + 14 = 12 - 8x$
	3. $7k = 3k - 36$	**4.** $12 - 2u = 9u + 45$
	5. $11 - m = 51 - 6m$	**6.** $-10k + 1 = 40 - 7k$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	7. $3(6p - 1) = 11p - 45$	**8.** $2(4w - 1) = -10(w - 3) + 4$
	9. $9(m - 3) + 3m = 7m + 43$	**10.** $-16 - 7(2a + 3) = 23 - 2a$
	11. $5x - (x - 18) = 6 - 2(x + 15)$	**12.** $8(y + 4) - 2(y - 1) = 70 - 3y$

Summary: _____

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 2:** Variables on Both Sides

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $4x + 1 = 2x - 5$

2. $y - 3 = -3y - 43$

3. $7 - 14m = 2m - 5$

4. $-10b + 5 = 7b + 5$

5. $r + 15 = 4r - 6$

6. $10 - 2v = -5v - 50$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

7. $3(5y + 1) = 18y$	**8.** $4 - 3x = 8 - 4x$
9. $8(2w - 2) = 7(3w + 2)$	**10.** $3(5x + 2) = 2(3x - 6)$
11. $6(m - 1) = 3(3m + 5)$	**12.** $7(p + 3) + 9 = 5(p - 2) - 3p$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

NO SOLUTION & INFINITE SOLUTION

No Solution	Infinite Solution
$-4(2x + 1) = -8x - 2$	$-5 - 9x = 3(1 - 3x) - 8$
There is **no possible number** that could work as a solution to the equation!	**Every number** could work as a solution!

MORE EXAMPLES!

1. $3(2x + 2) - 3x = 6 + 3x$

2. $6(2x - 6) = -7(-2x + 4)$

3. $8(5x - 3) = 6(-3x - 4)$

4. $3x - 13 = 7(x + 2) - 4(x - 7)$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 3:** No Solution/Infinite Solution

Directions: Solve each equation. **SHOW ALL STEPS!**	
1. $2(n + 2) = 4n + 1 - 2n$	**2.** $-15 - 3w = 4w - 2w$
3. $2(x + 4) = 4x + 3 - 2x + 5$	**4.** $4(2r - 1) = -2(3r + 16)$
5. $3a - 6a + 2 = 8a + 20 - 5a$	**6.** $-(v + 4) + 5 = 4v + 1 - 5v$
7. $8 - 6m = 4m + 48$	**8.** $11 + 3p - 7 = 6p + 5 - 3p$
9. $4y = 38 - \frac{1}{2}(4y + 16)$	**10.** $8x + 11 = 2(4x - 7) + 25$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
STEPS TO SOLVE	
GUIDED EXAMPLES	1) $\frac{4}{3}=\frac{x}{6}$ 2) $\frac{2}{10}=\frac{4}{n+5}$ 3) $\frac{y-3}{2}=\frac{y}{5}$ 4) $\frac{2k-3}{k+9}=\frac{3}{5}$
YOU TRY!	**Directions:** Solve each proportion. **1.** $\frac{x+10}{8}=\frac{x}{3}$ **2.** $\frac{p+2}{p+7}=\frac{3}{6}$ **3.** $\frac{w-2}{w}=\frac{3}{6}$ **4.** $\frac{n-4}{6}=\frac{n-7}{9}$ **5.** $\frac{3}{2}=\frac{a-3}{a+1}$ **6.** $\frac{25}{f+2}=\frac{5}{3}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

	7. $\frac{8}{m-9}=\frac{6}{m-8}$	**8.** $\frac{2}{r+4}=\frac{4}{r}$
	9. $\frac{5}{2}=\frac{v-8}{v-2}$	**10.** $\frac{6}{k-1}=\frac{8}{k-6}$
	11. $\frac{y-7}{y-9}=\frac{3}{4}$	**12.** $\frac{h-6}{9}=\frac{h+9}{4}$
	13. $\frac{r}{2}=\frac{r+10}{4}$	**14.** $\frac{7}{2x-8}=\frac{3}{x-6}$
	15. $\frac{2}{3u+7}=\frac{3}{4u+9}$	**16.** $\frac{2x-1}{5}=\frac{3x+1}{10}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 4:** Proportions

Directions: Solve each equation. **SHOW ALL STEPS!**	
1. $\frac{x}{4}=\frac{3}{2}$	**2.** $\frac{f+3}{12}=\frac{7}{2}$
3. $\frac{4}{n+9}=\frac{2}{7}$	**4.** $\frac{3}{8}=\frac{s-2}{24}$
5. $\frac{3}{10}=\frac{b+7}{20}$	**6.** $\frac{3}{16y}=\frac{1}{y+13}$
7. $\frac{m}{3}=\frac{m+4}{7}$	**8.** $\frac{a}{2}=\frac{a-4}{6}$
9. $\frac{x+2}{3}=\frac{x+4}{5}$	**10.** $\frac{n+8}{5}=\frac{n-1}{2}$
11. $\frac{p-5}{p+1}=\frac{2}{3}$	**12.** $\frac{x+2}{6}=\frac{2x-10}{8}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Properties of Equality

The steps to solve an equation are justified by properties of equality. They ensure that the equation stays balanced in order to solve for the missing variable.

- Addition Property of Equality
- Subtraction Property of Equality
- Multiplication Property of Equality
- Division Property of Equality
- Distributive Property

Directions: Identify which property justifies each step used to solve the equations below.

1

Equation Steps	Properties/Reasons
1. $2(x-9) = -10$	**1.** Given
2. $2x - 18 = -10$	**2.**
3. $2x = 8$	**3.**
4. $x = 4$	**4.**

2

Equation Steps	Properties/Reasons
1. $\frac{m}{-3} + 10 = -1$	**1.** Given
2. $\frac{m}{-3} = -11$	**2.**
3. $m = 33$	**3.**

3

Equation Steps	Properties/Reasons
1. $\frac{5y-1}{2} = 7$	**1.** Given
2. $5y - 1 = 14$	**2.**
3. $5y = 15$	**3.**
4. $y = 3$	**4.**

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

4

Equation Steps	Properties/Reasons
1. $10k - 4 = 2k - 20$	**1.** Given
2. $8k - 4 = -20$	**2.**
3. $8k = -16$	**3.**
4. $k = -2$	**4.**

5

Equation Steps	Properties/Reasons
1. $5n - 42 = 12n$	**1.** Given
2. $-42 = 7n$	**2.**
3. $-6 = n$	**3.**
4. $n = -6$	**4.**

6

Equation Steps	Properties/Reasons
1. $-8(w + 1) = -5(w + 10)$	**1.** Given
2. $-8w - 8 = -5w - 50$	**2.**
3. $-3w - 8 = -50$	**3.**
4. $-3w = -42$	**4.**
5. $w = 14$	**5.**

7

Equation Steps	Properties/Reasons
1. $14 - 2(x + 8) = 5x - (3x - 34)$	**1.** Given
2. $14 - 2x - 16 = 5x - 3x + 34$	**2.**
3. $-2x - 2 = 2x + 34$	**3.**
4. $-4x - 2 = 34$	**4.**
5. $-4x = 36$	**5.**
6. $x = -9$	**6.**

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Steps to Solve		Make sure the absolute value expression is _____.
		Set the "inside" equal to both the positive and negative number from the other side of the equation.
		_____ both equations.
		Write your answer as a _____ _____. $x = \{__, __\}$
Examples	**Directions:** Solve each equation. Write your answer as a solution set.	
	1. $\|x\| = 2$	**2.** $\|m\| = 14$
	3. $\|5z\| = 40$	**4.** $\|-7a\| = 28$
	5. $\|d + 1\| = 8$	**6.** $\|w - 3\| = 4$
	7. $\|4n + 2\| = 34$	**8.** $\|-2v - 5\| = 17$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

What if there is "stuff" outside the Absolute Value Bars?

9. $|c| + 2 = 12$

10. $|y| - 8 = -5$

11. $6|x| = 24$

12. $-2|z + 3| = -14$

13. $\frac{|2w - 4|}{6} = 3$

14. $-10|h + 5| - 3 = -83$

15. $7|4n + 8| - 6 = 106$

16. $\frac{3}{2}|4r - 4| - 24 = -21$

Special Cases

Directions: Solve the equations below. Explain their solutions.

17. $|x| = -6$

18. $-3|2x + 1| = 21$

Explain:

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 5:** Absolute Value Equations

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $\lvert x \rvert = 3$	**2.** $\lvert m \rvert = -10$
3. $\lvert -6k \rvert = 30$	**4.** $\lvert c - 10 \rvert = 3$
5. $\lvert 2k + 3 \rvert = 7$	**6.** $\lvert -4g - 2 \rvert = 10$
7. $\left\lvert \frac{1}{2}x + 5 \right\rvert = 13$	**8.** $\lvert 26 - 4c \rvert = -2$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

9. $-3\lvert p\rvert = 12$	**10.** $\lvert m\rvert + 2 = 11$
11. $\dfrac{\lvert 4x - 12\rvert}{2} = 8$	**12.** $\lvert 3b + 9\rvert - 16 = -52$
13. $\dfrac{4}{3}\lvert 3w - 6\rvert = 16$	**14.** $-5\lvert 6a + 21\rvert = -15$
15. $4\lvert v - 2\rvert - 3 = 25$	**16.** $3 - \dfrac{1}{2}\lvert r + 9\rvert = -11$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Literal Equations

SOLVE EACH OF THE EQUATIONS BELOW FOR x:

$2x - 5 = 13$	$ax - b = c$

Quick Write



Identify the similarities and differences between the two equations:

One-Step Problems

1. $A = lw$ solve for w	**2.** $A = bh$ solve for h
3. $d = rt$ solve for t	**4.** $I = prt$ solve for t
5. $V = lwh$ solve for h	**6.** $A = \pi r^2$ solve for r
7. $m = c - s$ solve for c	**8.** $D = \frac{m}{v}$ solve for m

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Multi-Step Problems

Helpful Hints:

- Think backwards ← PEMDAS
- Remove fractions by multiplying by the reciprocal.
- Last step is USUALLY to divide by whatever is next to your variable.

9. $A = \frac{1}{2}bh$ solve for h

10. $V = \frac{1}{3}Bh$ solve for B

11. $K = \frac{mv^2}{2}$ solve for m

12. $a = \frac{b+c}{d}$ solve for b

13. $P = 2L + 2W$ solve for W

14. $Ax + By = C$ solve for y

15. $y = mx + b$ solve for x

16. $A = P + Prt$ solve for t

17. $C = \frac{5}{9}(F - 32)$ solve for F

18. $A = \frac{1}{2}h(b_1 + b_2)$ solve for b_1

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 7:** Equations Review

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $5(n + 4) - (n + 7) = -11$	**2.** $2(x - 8) + 4x = 6(x - 2) - 4$
3. $2(k + 4) = 3(k - 2)$	**4.** $17 - 3(w + 5) = 6(w + 5) - 2w$
5. $9y - (2y - 3) = 5(y - 2) + 2y$	**6.** $-8(a - 4) + 3a = 2(4a + 9) + 1$
7. $\frac{n-6}{3} = \frac{n-2}{9}$	**8.** $\frac{10}{2x+4} = \frac{4}{x+5}$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

9. $\|x + 9\| = 7$	**10.** $\|2m - 1\| = 3$
11. $\|3g\| - 4 = 11$	**12.** $3\|2v - 7\| - 4 = -1$
13. Solve $A = \frac{1}{2}bh$ for b	**14.** Solve $P = 2l + 2w$ for l
15. Solve $s = n(a + 1)$ for a	**16.** Solve $\frac{1}{3}(2x - y) = z$ for x

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Using Algebra to Solve Word Problems

1. **DEFINE A VARIABLE**
   Use **"LET STATEMENTS"** to define your variable.

2. **SET UP EQUATION & SOLVE**
   Translate into an equation using your let statements. Then solve!

3. **DEFINE ANSWER**
   Give exactly what the problem is asking for.

Set I: Finding Two Numbers

1. The larger of two numbers is four more than the smaller number. If the sum of the numbers is 74, find the numbers.

2. The larger of two numbers is seven less than three times the smaller number. If the sum of the numbers is 61, find the numbers.

3. The larger of two numbers is one more than four times the smaller number. If the sum of the numbers is 106, find the numbers.

4. The larger of two numbers is twice the smaller number. If 12 is subtracted from the larger number, the result is 7 more than the smaller number. Find the numbers.

5. The highest score on an Algebra test was 42 points more than the lowest. When added together, the lowest and highest score was 154. Find both the highest and lowest score.

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

6. Garrett has one dollar less than three times as much money as Liz has. Together they have $179. How much money does Garrett have?

7. The Buffalo Bills scored 24 more than twice the number of points that the Miami Dolphins scored. Altogether, the teams scored 66 points. How many points did each team score individually?

8. There are 10 less red skittles than orange skittles in the bag. The orange skittles are also twice the number of red skittles. Find the number of red and orange skittles in the bag.

Set 2: *Perimeter of a Rectangle*

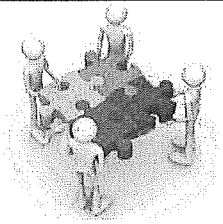
9. The length of a rectangle is six inches more than its width. If the perimeter of the rectangle is 24 inches, find its dimensions.

10. The length of a rectangle is five inches more than four times its width. If the perimeter of the rectangle is 90 inches, find its dimensions.

11. The length of a rectangle is three centimeters less than twice its width. If the perimeter of the rectangle is 18 centimeters, find its dimensions.

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

Group Members:	Per: _____



REVIEW: *Equations & Word Problems*

Directions: Work together to solve each problem. **SHOW ALL WORK!** Each person should be participating. At the end of class, one person's paper will be chosen at random and graded for the group.

Directions: Solve each equation. **SHOW ALL STEPS!**	
1. $-2(4x + 3) - (x - 7) = 46$	**2.** $-\frac{4}{3}(12k + 27) = -57 - 9k$
3. $23 - 2(5a + 9) = 5(2a - 11)$	**4.** $\frac{5}{2}(6w - 16) = 18w - (3w + 40)$
5. $\frac{10}{a-6} = \frac{5}{6}$	**6.** $\frac{g}{8} = \frac{g-1}{10}$
7. $\frac{z-9}{6} = \frac{z+7}{3}$	**8.** $\frac{5}{2} = \frac{4x+21}{x-3}$
9. $\lvert x \rvert = 14$	**10.** $\lvert 8m - 8 \rvert = 40$

© Gina Wilson (All Things Algebra®, LLC), 2012-2016

11. $\left\|\frac{m}{7}\right\| + 4 = 5$	**12.** $-5\|4p + 4\| = -40$
13. Solve $a = \frac{v}{t}$ for v.	**14.** Solve $-8 = \frac{1}{4}(m - n)$ for m.
15. Solve $a^2 + b^2 = c^2$ for b.	**16.** Solve $S = 2\pi r(h + r)$ for h.

Directions: Use the **3-step process** to solve each word problem.

17. The larger of two numbers is two more than five times the smaller number. If the sum of the two numbers is 80, find both numbers.

18. Find three consecutive odd integers such that the sum of the smallest number and middle number is 27 less than 3 times the largest number.

19. The length of a flat screen TV is 5 inches less than 3 times its width. If the perimeter of the TV is 150 inches, find the dimensions.

20. Olivia and Camille went clothes shopping. Camille spent $19.40 less than twice what Olivia spent. In total, they spent $272.35. Find the amount each girl spent.

© Gina Wilson (All Things Algebra®, LLC), 2012-2016